

# PREDICTING APPENDIX CANCER RISK USING MACHINE LEARNING

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## 1. Dataset

**Appendix Cancer Prediction Dataset (Kaggle) –**

**<https://www.kaggle.com/datasets/ankushpanday1/appendix-cancer-prediction-dataset>**

A patient-level healthcare dataset containing 260,000 records across 25 countries (including India, with 35,350 records). It includes demographic details (Age, Gender, BMI, Country), lifestyle factors (Smoking, Alcohol Consumption, Physical Activity, Diet), medical history (Family History of Cancer, Genetic Mutations, Chronic Diseases, Previous Cancers), clinical measurements (Blood Pressure, Cholesterol, WBC/RBC/Platelet Counts, Tumor Markers), and diagnosis details (Symptom Severity, Diagnosis Delay, Treatment Type, Survival Years). The target column, Appendix\_Cancer\_Prediction, labels each patient as cancer-positive (Yes) or cancer-negative (No).

## 2. Problem Statement

*Given a patient's demographic, lifestyle, and clinical profile, can a machine learning model accurately predict whether the patient will be diagnosed with appendix cancer (Yes/No)? A secondary goal is to identify which risk factors contribute most strongly to this prediction, so that high-risk patients can potentially be flagged earlier for screening.*

## 3. Proposed Solution

The problem will be addressed as a binary classification task using a standard supervised machine learning pipeline:

- Clean the data – handle missing values in Chronic\_Diseases and Treatment\_Type.
- Encode categorical features (Gender, Smoking\_Status, Diet\_Type, etc.) and scale numeric features where required.
- Split the data into training and test sets (80/20) using stratified sampling to preserve class balance.
- Train classification models to predict Appendix\_Cancer\_Prediction, applying class-weighting to handle the imbalance between cancer-positive and cancer-negative cases.
- Evaluate models using Precision, Recall, F1-score, and ROC-AUC (not accuracy alone, since the target is imbalanced).
- Use feature importance from the trained model to identify and report the top risk factors driving predictions.

## 4. Machine Learning Algorithms

| Algorithm                | Role & Justification                                                                                                                                                                  |
|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Logistic Regression      | Baseline model. Simple and interpretable; establishes a reference performance level before trying more complex models.                                                                |
| Random Forest Classifier | Primary model. Handles a mix of categorical and numeric features well, is robust to outliers in clinical values, and provides feature importance scores to identify key risk factors. |